

Analysis and Optimization of DSP Algorithms for fast convergence and accurate mitigation of optical fibre impairments

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I. INTRODUCTION AND PROBLEM STATEMENT

Fiber optical communication systems exhibit a range of receiver impairments such as phase offset, frequency offset, crosstalk, dispersion, and other channel distortions. These impairments are often time-varying in nature, and in order to establish communication, it is necessary that these impairments are estimated and corrected for. Although adaptive equalizers to correct for each of these impairments exist, there is a lot of scope to improve their speed of convergence. Also, these algorithms fail for denser modulation schemes, like 64-QAM and 128-QAM. The goal of this project is to come up with DSP-based algorithms that can achieve faster convergence and more accurate impairment mitigation, using the presently commonly used algorithms as a benchmark.

II. THEORY OF DIFFERENT IMPAIRMENTS

The below section summarizes the cause of different impairments in optical fibers and the existing methods for correction, along with how these impairments are modeled. The main impairments considered in this project are polarization effects, frequency offset, phase noise, and crosstalk. Dispersion is another impairment that has considerable effect, but this has not been yet incorporated into our system.

A. Polarization Effects

Single mode fibers support the propagation of 2 orthogonally polarized modes, which ideally have the same propagation constant and are degenerate. However, the non perfect circularity of the core and cladding cause a loss of degeneracy which leads to pulse distortions at the receiver's end. The fiber acquires birefringence, which is a difference between the refractive indices of the 2 polarization modes. This causes a temporal spread of pulses, called polarization mode dispersion (PMD). Pulses in orthogonal polarizations acquire distinct group velocities, and a mutual delay called the differential group delay (DGD) develops between projections of the linearly polarized signal onto its orthogonal states, given by

$$\tau = \frac{l\Delta\beta}{\omega_0},$$

where l is the length, $\Delta\beta$ is the difference in propagation constants, and ω_0 is the angular carrier frequency.

This impairment is modeled by representing the electric field as a 2-d complex vector, and showing the transformation due to polarization by a frequency dependent matrix.

$$\begin{bmatrix} E_{o,x}(\omega) \\ E_{o,y}(\omega) \end{bmatrix} = H(\omega) \begin{bmatrix} E_{i,x}(\omega) \\ E_{i,y}(\omega) \end{bmatrix}.$$

A single birefringent section is modeled as

$$H^{(i)}(\omega) = V^{(i)} \Lambda^{(i)}(\omega) U^{(i)H},$$

where $U^{(i)}$ and $V^{(i)}$ are 2×2 unitary matrices representing random polarization coupling, and

$$\Lambda^{(i)}(\omega) = \begin{bmatrix} e^{j\omega\tau^{(i)}/2} & 0 \\ 0 & e^{-j\omega\tau^{(i)}/2} \end{bmatrix}$$

incorporates the DGD $\tau^{(i)}$ of that segment. The coupling matrices are generated by applying singular value decomposition (SVD) to random complex Gaussian matrices.

Long fibers are modeled as a cascade of N such birefringent sections, each with different random polarization coupling unitary matrices. The overall transfer function of the fiber is then the product of the individual matrices across sections.

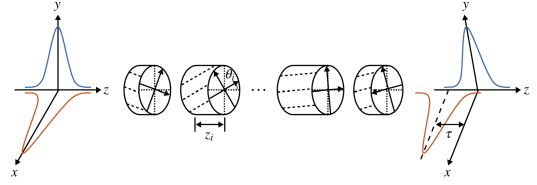


Fig. 1. Effect of polarization mixing across sections of fiber

Because polarization effects fluctuate stochastically in real-world deployments due to different environmental changes, coherent receivers dynamically correct for them, employing adaptive polarization-tracking equalizers to mitigate PMD-induced distortions.

B. Frequency Offset

Frequency offset occurs when the transmitter and receiver oscillators have a difference in frequencies, that causes a phase rotation of the received signal. The impairment is modeled as the multiplication of the received samples by $e^{j2\pi\Delta f n T_s}$, where Δf is the carrier frequency mismatch and T_s is the sampling interval. A method for rectifying this in the case of QPSK involves raising the received signal to the fourth power such that the modulation is removed and the resulting phase contains only the term $4(2\pi\Delta f n T_s)$. Taking the unwrapped phase and dividing by four provides an estimate of the frequency offset, which can be removed.

C. Phase Noise

Phase noise arises from random temporal fluctuations in the phase of the optical carrier. These fluctuations are typically modeled as $\phi[k] = \phi[k-1] + \Delta\phi[k]$, with $\Delta\phi[k]$ being a zero-mean Gaussian variable whose variance is proportional to the combined laser linewidth. Phase noise causes constellation rotation and degradation in coherent detection. Phase noise can be corrected using the LMS algorithm below to track and compensate for phase drift.

III. SYSTEM MODEL: EXISTING ALGORITHMS FOR MITIGATION

LMS (Least Mean Squares) Algorithm: The LMS algorithm is an adaptive filtering method that minimizes the mean squared error between the received signal and the transmitted signal, which is then used to update the filter coefficients via stochastic gradient descent.

Constant Modulus Algorithm (CMA): CMA relies on the constant modulus property of certain modulations such as QPSK, where the transmitted signal has a fixed amplitude. The key idea is that ISI introduces variations in the amplitude of the received signal, and by enforcing a constant amplitude at the equalizer output, CMA reduces ISI. CMA defines a cost function based on the deviation of the output signal's modulus from a reference dispersion constant R_d , which is then used for stochastic gradient descent:

$$J_{\text{CMA}} = \mathbb{E} [(R_d - |y[k]|^2)^2]$$

The equivalent error used for gradient descent is:

$$e_{\text{CMA}}[k] = (R_d - |y[k]|^2) y[k]$$

The filter coefficients are updated as:

$$w[n+1] = w[n] + \mu x[n] e_{\text{CMA}}^*[k]$$

CMA works best with constant modulus signals like in QPSK, but it can handle QAM reasonably well as well. There are also versions of the algorithm which involve mapping it onto one of 3 radii rather than a single fixed one, which takes non-constant amplitudes into account. One point to note here is that CMA corrects only the modulus- the phase offset will need to be corrected separately.

IV. OVERVIEW OF SOLUTION APPROACH

The following is a summary of the different experiments carried out for the analysis and optimization of the above algorithms:

- 1) Starting with the standard CMA algorithm for QPSK constellations impaired with polarization mixing, delay, and AWG noise, and then implementing different modifications: Momentum added CMA, variable momentum added CMA, LBFSGS (Second order based CMA), and first and second moment based CMA
- 2) Comparing rates of convergence of these different algorithms to standard CMA in terms of convergence speed and EVM of converged symbols
- 3) Implementing MMA and CMA, and comparing the two in terms of speed of convergence and accuracy

- 4) Observing trends between different parameters and validating these: EVM versus number of filter taps, convergence speed versus input SNR, convergence speed versus number of filter taps, convergence symbol versus differential group delay, etc.
- 5) Adding CFO as an additional impairment along with PMD and AWGN for both QPSK and 16-QAM
- 6) Devising an algorithm to fix all 3 impairments with a single system (as opposed to fixing PMD with CMA followed by fixing CFO with 4th order correction).
- 7) Shifting from blind to pilot-assisted correction, finding an efficient algorithm for pilot synchronization

V. DETAILED SOLUTION IMPLEMENTATION

A. Problem Set-up

We generate 10^6 constellation symbols, following either QPSK or 16-QAM modulation (both have been implemented). We further implement polarization mixing and a delay of $19.76 \text{ ps}/\sqrt{\text{km}}$, which translates to about a 2 symbol delay for a data-rate of 32 Gbaud across a 10 km long fiber split into 20 sections. AWGN is also added to these symbols, whose SNR can be adjusted as a parameter.

Frequency offset is also added as an impairment for the later experiments- the symbols are offset by a frequency of 2 kHz.

The problem is modeled as a search for the optimal values of 51 filter taps that should compensate for these impairments.

B. Algorithms Implemented

1) **CMA:** The standard CMA is described in the previous section. CMA iteratively adjusts the parameters based on the 1st derivative (gradient) of the cost function to minimize error.

2) **Momentum-CMA (MCMA):** MCMA accelerates convergence by introducing an additional momentum term that is proportional to the previous tap weight increment. The filter taps are updated according to the following equation:

$$g(k+1) = g(k) + \mu \nabla J(g(k)) + \alpha [g(k) - g(k-1)]$$

where α is the momentum factor ($\alpha = 0.8$ in our case).

3) **Variable Momentum CMA (VMCMA):** VMCMA adapts the momentum factor dynamically, reducing it as convergence is approached. The tap update rule remains similar to MCMA:

$$g(k+1) = g(k) + \mu \nabla J(g(k)) + \alpha(k) [g(k) - g(k-1)]$$

However, the momentum term $\alpha(k)$ is updated dynamically:

$$\alpha(k) = \eta \alpha(k-1) + \gamma |\nabla J(g(k))|^2$$

taking an update proportional to the magnitude of the gradient, since this value becomes smaller as convergence is approached. For implementation, the parameters were set to $\eta = 0.95$ and $\gamma = 1 \times 10^{-3}$, with α initialized at 0.8 and clipped to remain within the range of $[0.1, 0.95]$.

4) *Limited Memory BFGS Algorithm (LBFGS)*: The LBFGS algorithm utilizes both the first and second derivatives to make more accurate directed steps during optimization. Because calculating the full inverse Hessian matrix for the second derivative is computationally expensive, L-BFGS uses a lower-rank approximation based on a limited memory of past update values (configured to $m = 10$).

Unlike CMA's symbol-by-symbol updates, L-BFGS operates on an entire block of N symbols at once, computing the gradient update and CMA error as the loss across the full block, and performing multiple iterations per block.

5) *2nd Moment CMA*: The 2nd Moment CMA utilizes an Adam-like optimization scheme, incorporating the first moment (moving average of the gradients) and the second moment (variance of the gradients). The moments are estimated using decay factors β_1 and β_2 :

$$m_i = \beta_1 \cdot m_{i-1} + (1 - \beta_1) \cdot g_i$$

$$v_i = \beta_2 \cdot v_{i-1} + (1 - \beta_2) \cdot g_i^2$$

To account for initialization bias, bias-corrected estimates are calculated based on the number of completed iterations (t):

$$\hat{m}_i = \frac{m_i}{1 - \beta_1^t} \quad \hat{v}_i = \frac{v_i}{1 - \beta_2^t}$$

The filter parameters p are then updated using these corrected moments:

$$p(k+1) = p(k) + \mu \frac{\hat{m}(k)}{\sqrt{\hat{v}(k)} + \epsilon}$$

The algorithm was implemented with a learning rate $\mu = 0.5 \times 10^{-4}$, moment decay factors $\beta_1 = 0.8$ and $\beta_2 = 0.8$, and a stability constant $\epsilon = 1 \times 10^{-8}$.

C. Performance of these algorithms

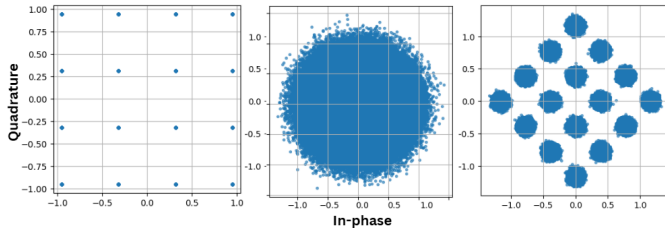


Fig. 2. a) 10^6 ideal symbols in 16-QAM modulation, b) 16-QAM symbols impaired with PMD, CFO and noise, c) Recovered symbols with CMA (residual rotation since CMA is phase-blind)

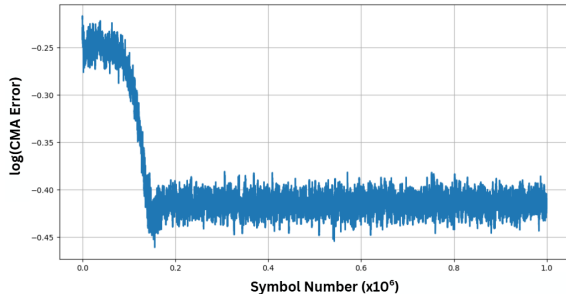
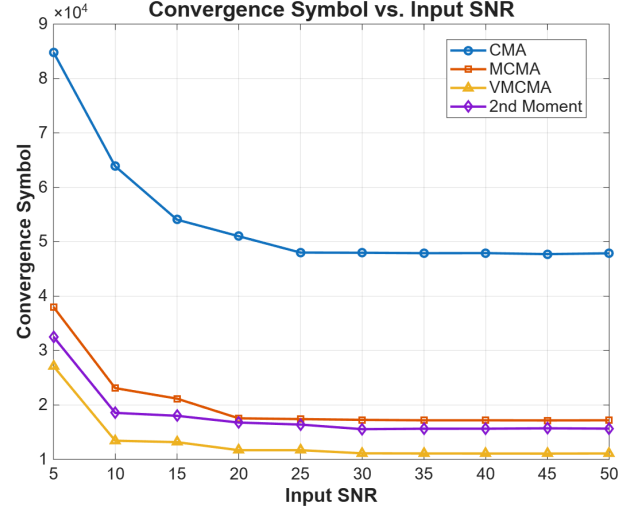


Fig. 3. $\log(\text{CMA Error})$ vs symbol index to identify convergence symbol



These results are for 16-QAM, impaired with polarization mixing, having a group delay equivalent to 2 symbols, and AWGN of varying input SNR. The convergence is evaluated by setting a threshold based on the mean and variance of the converged part of the sequence. We see faster convergence for higher SNRs, and we can also clearly see the improvement in convergence across the different algorithms. VMCMA performs the best, followed by 2nd moment CMA, MCMA, and then the standard CMA taking almost 50000 more symbols to converge.

L-BFGS is different from all the above algorithms in the sense that it operates over the entire block of symbols at once. So, while the MCMA converges to an error of about 0.03 after roughly 20000 symbol passes, L-BFGS gives an average error of ≈ 0.031 across a 2500 symbol block. The number of symbol passes here, however, is now the number of symbols, say 2500, times the number of iterations (set to a maximum 200), which is about 25 times the processing time of CMA for the same E_x and E_y sequences. Further optimization has to be carried out.

D. Comparing Multi-Modulus Algorithm (MMA) to CMA

MMA is another algorithm that is similar to CMA, except that it maps constellations onto one of several radii, rather than a single average one like in CMA. While CMA initially provides robust blind equalization, its single-radius cost function introduces significant mismatch for higher-order schemes like 16-QAM, where symbols occupy multiple distinct amplitude levels. MMA exhibits a slower convergence as compared to CMA because of this multi-level cost surface, but it shows significantly higher accuracy, since it models the true geometry of the modulation scheme, which is not the case in CMA. We use Error Vector Magnitude as a metric to measure the accuracy of recovery: EVM measures the deviation of the measured symbol from its ideal target position, represented as an error vector. It is expressed as:

$$\text{EVM} = \sqrt{\frac{1}{N} \sum_{k=1}^N |S_{\text{ideal},k} - S_{\text{actual},k}|^2}$$

where N is the number of symbols, $S_{\text{ideal},k}$ is the ideal constellation point and $S_{\text{actual},k}$ is the measured symbol. The performance of MMA versus CMA can be seen in the graphs below- MMA achieves a much lower residual EVM, but takes much longer to converge.

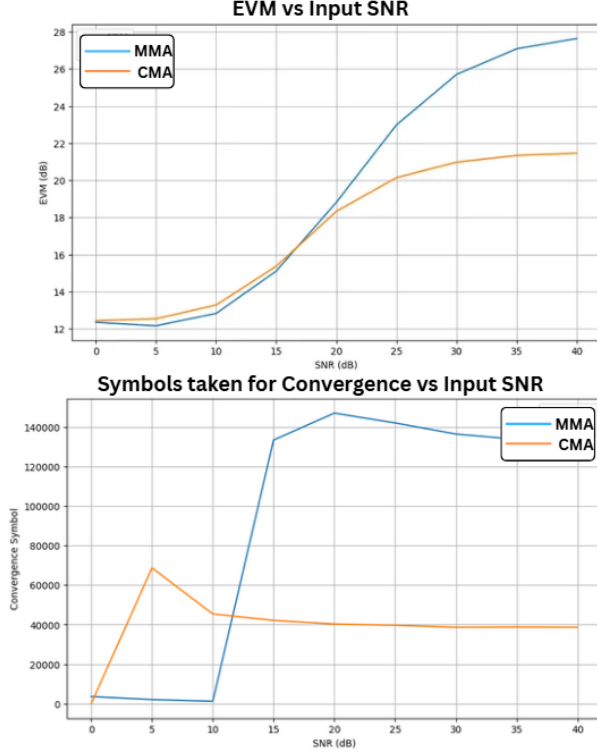
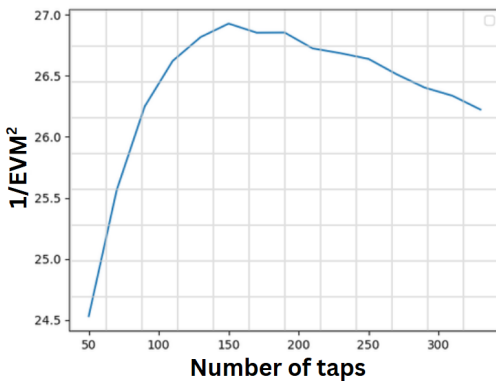


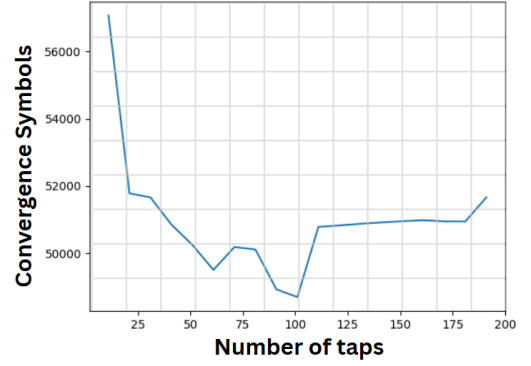
Fig. 4. a) MMA lower error than CMA, b) MMA takes more symbols to converge

E. Observing Different Trends

1) *EVM versus number of taps*: EVM is maximum for some ideal number of taps, and then deteriorates as the number of taps increases even further.

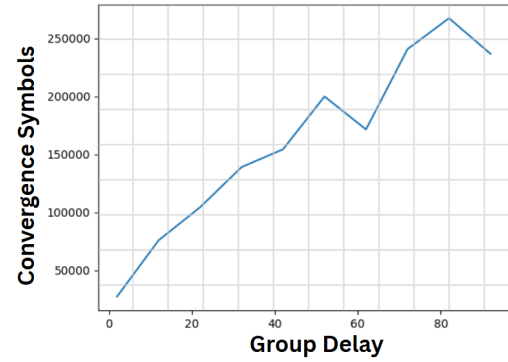


2) *Convergence speed versus number of taps*: We do not find any particular correlation between speed of convergence and number of taps- the number of symbols taken to converge is approximately the same for a varying number of taps.



3) *Convergence speed versus input SNR*: This trend can be seen in the plot shown in the previous section comparing the different algorithms- the convergence being achieved faster for high input SNR symbols, since there is less noise to compensate for.

4) *Convergence speed versus group delay*: As expected, as group delay increases, the symbols get more and more distorted, and it takes longer to recover the ideal constellations.



F. Compensation for frequency offset, polarization mixing and group delay, and noise

Currently, the process to compensate for PMD, CFO and noise entails carrying out LMS or CMA to correct PMD and noise, and then use 4th order correction to correct CFO (carrier frequency offset).

By raising the received symbols to the fourth power, the inherent modulation phases are effectively eliminated. The ideal phase states of these symbols are odd multiples of $\pi/4$, so multiplying their phase by 4 maps them to a uniform phase alignment. The CFO can then be extracted by locating the spectral peak using FFT, which can be used to rotate and correct the symbol stream.

The new algorithm we propose implements a joint, decision-directed LMS approach for simultaneous channel equalization (for PMD and noise correction) and CFO estimation. Iteratively, the received baseband signal is convolved with the current estimated equalizer filter taps, and subsequently phase-rotated by the current CFO estimate to perform active frequency correction. The loss function used for the gradient descent that drives the filter tap and frequency estimate updates

are found by mapping the received signals to the nearest ideal constellation point, and calculating the distance between them.

This algorithm showed good results, estimating the frequency offset within an error of 5%. The convergence takes longer to achieve as expected, but we are able to achieve finer and more accurate CFO correction through this, in addition to the added benefit of using a single system for both impairments without worrying about the added computational complexity of taking FFT in the 4th order correction method.

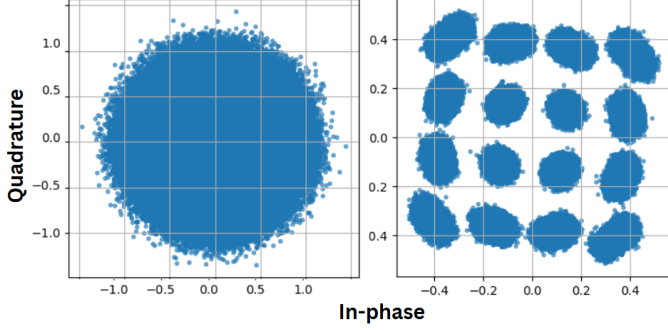


Fig. 5. a) 16-QAM impaired with PMD, CFO and noise, b) Corrected 16-QAM symbols using proposed joint LMS algorithm

G. Moving from blind to pilot-assisted equalization

In reality, we are not entirely in the dark about the symbols being transmitted. We have pilot symbols, which are predetermined, known reference signals periodically embedded within the transmitted data. By comparing the received pilots against their ideal reference, the receiver can use this information to estimate impairments, perform equalization, and track frequency and phase offsets.

However, the effectiveness of pilot-aided algorithms is entirely dependent on precise frame synchronization- we do not know which bit the pilots begin at, i.e., there is a need to identify the exact start offset (τ) of the incoming data frame. If the receiver is misaligned by even 1 symbol, the error metric will blow up, since it will then be computed between a random payload and the expected pilot sequence.

The initial implementation of the synchronization and equalization algorithm uses a structured pilot framing scheme for QPSK signals, where each 3172-symbol frame is divided into 116 sections of 32 symbols each. Known pilot symbols, generated via a linear feedback shift register (LFSR) polynomial, occupy the first 11 positions of the frame, and the first position of all subsequent sections.

To synchronize the incoming data stream, an unknown frame offset, denoted as τ , is initially estimated using a naive cross-correlation search. This estimated τ initializes a modified LMS algorithm that dynamically switches its cost function: it uses a data-aided mode to compute the error against the exact sequence at known pilot indices, and a decision-directed mode to compute the error against the nearest ideal constellation point for the rest of the data. The system operates in an iterative loop, utilizing the equalized CFO-corrected output symbols to make a better re-estimate of τ . This cycle of equalization and frame syncing repeats until the offset τ , the

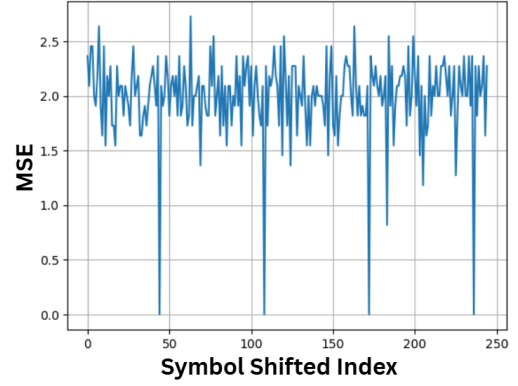


Fig. 6. MSE between shifted masked symbols and pilot sequence plotted against shift index- MSE=0 at periods of frame length, and random otherwise

complex filter taps, and the frequency estimate all converge. We find that with this algorithm, the symbols converge more accurately than it does without using the information from the pilots, but it takes about the same time.

Configuration	Symbols for Convergence	CFO Estimate Error
With Pilots	132187	3.4 %
Without Pilots	128432	8.3 %

TABLE I
COMPARISON OF CONVERGENCE AND CFO ESTIMATION ACCURACY WITH AND WITHOUT PILOT SYMBOLS.

VI. CONCLUSION AND NEXT STEPS

A summary of steps taken:

This project has delved into optimizing the LMS and CMA algorithms for mitigating noise, CFO and polarization mixing with delay for QPSK and 16-QAM constellations. We have implemented different optimizations for LMS and CMA and compared their performance. We further analyzed the relations between accuracy and speed of convergence with number of taps of the adaptive equalization filter, the extent of noise, and the group delay in the input symbols. We have also successfully implemented a single algorithm which corrects all 3 impairments together, and compared the performance of this with and without the knowledge of pilot sequences. These steps serve as the basis for future algorithms, with our end goal being using the knowledge of one impairment correction to correct the other- ultimately improving the speed of convergence and accuracy for denser constellations as well.

Inferences and Insights:

- Highest speed of convergence achieved by the implemented VMCMA algorithm, reaching 8 times faster convergence than the standard CMA.
- CMA achieves a faster convergence than MMA, but MMA converges to a lower EVM, especially for denser QAM schemes.

- Increasing taps helps compensate for higher Differential Group Delay (DGD), but exceeding the optimal number degrades the accuracy.
- Convergence speed is higher for lower input SNRs.
- A Joint-LMS approach for simultaneous CFO and PMD correction is more computationally efficient and accurate than sequential processing of CMA followed by 4th-order FFT for CFO correction. This prevents residual PMD from degrading the performance of the CFO correction block.
- Moving from blind to pilot-assisted equalization improves CFO estimation (reducing error from 8.3 to 3.4 %, but takes about the same time to converge).

Communication theory tools used (keywords):

Adaptive filters for equalization, LMS and CMA algorithms, digital modulation schemes (QPSK, QAM), carrier frequency offset, polarization mixing, differential group delay, AWGN, convergence criterion and EVM, pilot synchronization using autocorrelation.

Conclusion

This concludes the work I have taken up this semester for undergraduate research. I would like to thank Prof. Lakshmi Narasimhan and Prof. Deepa Venkitesh for their guidance with this project, and also Charan Srikanth for the fruitful discussions and collaborations.